

SAHIL NIRANJAN

AI/ML Engineer | Production ML, LLM Systems & Evaluation
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Professional Summary

AI/ML engineer with three and a half years across financial services, healthcare, and higher education. **Started on mainframe batch extracts**, moved into **production ML for client risk workflows**, and now builds **LLM retrieval systems and the evaluation harnesses that gate them**. Same question every time - where does it fail, how is it measured, and which cases still need a person.

Technical Skills

Languages: Python, SQL, C++17, SAS, JCL (z/OS), Bash

LLM & GenAI: RAG, LangChain, LangGraph, embeddings, vector search, MCP servers, prompt engineering, human-in-the-loop routing

Evaluation & Reliability: eval harness design, context relevancy, faithfulness, answer relevancy, temporal validation, precision-recall for imbalanced labels, leakage and drift detection, failure analysis

Machine Learning: scikit-learn, XGBoost, pandas, NumPy, classification, forecasting, time series, imbalanced learning, feature engineering, hyperparameter tuning

Data & MLOps: Spark, Kafka, MLflow, Git, Docker, PostgreSQL, Power BI, PII/PHI handling, HIPAA

Work Experience

Bank of New York (BNY)

Jan 2026 - Apr 2026

AI/ML Engineer Intern

Boston, MA

- Cut **response latency 18%** and improved **retrieval speed 35%** across **120k+ business-critical financial and compliance documents** by reworking the search strategy inside a Python retrieval-augmented generation system built on **LangChain, LangGraph, embeddings, and vector search**.
- Settled which retrieval configuration shipped to production by building the **evaluation harness** that scored every candidate on **context relevancy, faithfulness, and answer relevancy** against a fixed question set.
- Kept confident wrong answers out of a regulated workflow by **routing low-confidence outputs to human review** and mapping **which question types it handled poorly**, through failure analysis with the **compliance and operations** users.

Northeastern University School of Law

Aug 2025 - Jun 2026

Graduate Senior Data Analyst

Boston, MA

- Built the department's analytics function from nothing as its **first technical hire** and moved roughly **80% of recommendations to implementation** by gathering requirements face to face and defining the metrics with administrators first.
- Recovered close to **8 staff hours per week** and cut reporting **cycle time about 18%** by migrating recurring manual reporting onto automated **SQL and Python pipelines** over confidential student records.
- Made the tooling survive handoff, still in **daily use** by staff who did not build it, by **putting access controls in the code rather than the process** and training non-technical users to run it themselves.

Wipro | Alight Solutions account

May 2023 - Aug 2024

Associate Analyst

India (US shift)

- Ran the day-to-day **JPMorgan Chase delivery stream** under the account manager, holding **multi-million-row payroll and benefits extracts** of **sensitive employee and claims data** to schedule through **Annual Enrollment peaks** with **JCL on z/OS, SQL, and SAS**.
- Cut recurring data discrepancies **38%** and turnaround **33%**, **from 12 days to 8**, by reconciling **full-file against change-only files**, generating **EDI 834 enrollment extracts**, and tracing failures to the source record through **job logs and SQL** before the delivery window closed.
- Moved into modeling after the first few months and put **anomaly detection and risk models** into production on pipelines carrying roughly **1.5M records per month**, using XGBoost and ensembles in Python, Spark, and Kafka with MLflow tracking.
- Caught **temporal leakage from random train/test splits** that was inflating reported model performance, and rebuilt evaluation around **time-based validation and precision-recall metrics** suited to imbalanced labels, so reported performance matched what the system actually did in production.

Capgemini

Jan 2022 - Apr 2023

Data Engineer

India

- Built and maintained the **Python and SQL data-processing pipelines** behind ML applications on **clinical records and utilization data**, preparing structured datasets, implementing validation checks, and resolving data issues ahead of model training and batch inference.
- Developed and evaluated **classification and forecasting models** with scikit-learn, pandas, and NumPy, covering feature engineering, model comparison, hyperparameter tuning, and performance analysis under senior guidance.
- Integrated model outputs with backend services and scheduled workflows and investigated production defects through logs and **root-cause analysis**.

Projects

LOB Predictor | C++17, Python, scikit-learn, CMake | [Code](#) | [Demo](#)

2026

- Held **end-to-end p99 latency under 0.7 ms** on commodity CPU by exporting an **18-feature L2 order-book model** to a flat weight file scored by a **hand-written C++17 engine** as a **single dot product, no ONNX or BLAS**.

S.T.A.R. Toolbox | Python, rules engine, regulatory data; capstone, code under NDA

2026

- Delivered a regulated-environment screening system end to end against real client deadlines, encoding the governing rules as deterministic logic so **every conclusion traces to the rule that produced it** and it **fails closed** on ambiguity.

Education

Master of Science in Analytics, Northeastern University, Boston, MA

July 2026

Bachelor of Technology in Computer Science and Engineering, GGSIPU, Delhi, India

Jun 2022

Certifications and Publications

Advanced Certificate in Data Science, IIIT Bangalore

October 2023

Patent (published application): Real-time 3D pose classification, Indian Patent Application No. 202211030731, 2022. | [Record](#)

Publication (peer-reviewed): Emotion classification via multi-modal deep learning, IJEAST Vol. 6, No. 7, 2021. | [Paper](#)